MALLIKARJUN JADI

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PROFESSIONAL SUMMARY

Final-year CSE student targeting ML/AI engineering roles. Built a from-scratch PyTorch CNN (ResNet + Squeeze-and-Excitation) reaching 97.58% / 97.90% test accuracy on RGB / 13-band Sentinel-2 imagery; fine-tuned Qwen2.5-1.5B with LoRA and QLoRA, lifting ROUGE-L from 0.107 to 0.356; and tuned a gradient-boosted Kaggle regression ensemble (0.118 RMSE). Supported by DSA fundamentals: 250+ LeetCode problems, 1745+ contest rating.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C
Machine Learning: Deep Learning, Computer Vision, CNNs, Residual Networks, Attention (Squeeze-and-Excitation), LLM Fine-Tuning (LoRA, QLoRA), Quantization, Model Evaluation, Hyperparameter Optimization, Ensemble Learning
Frameworks / Libraries: PyTorch, Torchvision, Hugging Face Transformers, PEFT, TRL, bitsandbytes, scikit-learn, XGBoost, CatBoost, Optuna, Pandas, NumPy, Matplotlib
Tools / Deployment: Hugging Face Hub, Streamlit, Gradio, Kaggle GPU, Git/GitHub
Full-Stack: MongoDB, Express.js, Node.js
DSA: 250+ LeetCode problems solved, competitive programming

SELECTED PROJECTS

EuroSAT Dual-Mode Land Cover Classifier | Computer Vision | GitHub, Live Demo

- Designed SEResEuroNet, a ResNet-style CNN with Squeeze-and-Excitation attention built from scratch in PyTorch (no pretrained backbones), reaching 97.58% test accuracy on RGB and 97.90% on 13-band Sentinel-2 imagery across 10 land-cover classes.
- Built a distribution-shift-safe pipeline: one frozen normalization-constants file shared by training and inference, and one modality-agnostic model class (in_channels = 3 or 13); used early stopping and adaptive LR scheduling with best-checkpoint restore.
- Deployed a dual-mode inference app on Streamlit with checkpoints hosted on Hugging Face Hub.

LoRA vs QLoRA Fine-Tuning of Qwen2.5-1.5B for Agricultural Q&A | LLM Fine-Tuning | GitHub, LoRA, QLoRA

- Fine-tuned Qwen2.5-1.5B-Instruct with LoRA (BF16, r=16) and QLoRA (4-bit NF4, double quantization) using PEFT and TRL SFTTrainer on Kaggle GPUs; 4,000 training / 500 validation / 500 test agricultural Q&A pairs with a leakage-safe split.
- LoRA raised ROUGE-L from 0.107 (base) to 0.356 and BERTScore F1 from 0.848 to 0.905; QLoRA reached 0.219 / 0.878. Fine-tuning cut mean answer length from 94.7 to 24.2 words (reference answers: 23.6).
- Found and fixed an evaluation bug (adapters sharing one model object contaminated Base/LoRA/QLoRA outputs) by rebuilding isolated-adapter evaluation with self-tests; documented factual-error failure cases and published both adapters on Hugging Face Hub.

House Prices: Advanced Regression (Kaggle) | Classical ML | GitHub

- Built a regression pipeline on 79-feature housing data with engineered features (house age, garage age, total area, bathrooms) and log1p skew correction.
- Tuned CatBoost and XGBoost with Optuna, validated with 10-fold CV, and blended them in an 80/20 weighted ensemble: 0.11799 RMSE (log-scale) on the Kaggle public leaderboard.

Fixmate: Multi-Role Service Booking Portal | Full-Stack | GitHub

- Built a MongoDB/Express/Node.js booking platform with 40+ service categories and admin, client, and worker dashboards; implemented all CRUD routes in vanilla JS with no third-party APIs.

LEADERSHIP & ACHIEVEMENTS

- First-time team lead for Smart India Hackathon after two earlier hackathon attempts were not shortlisted; also competed in HackFusion 1.0 and 2.0, Xypheria, and SrishtiTech 2026 (36-hour).
- LeetCode: 250+ problems, 1745+ contest rating (top 10.98%), 74-day max streak, 86.62% submission accuracy.

EDUCATION

B.E. in Computer Science & Engineering, AGMR College of Engineering & Technology, Hubli *2023 – 2027* | *CGPA 8.0/10 (till 6th sem)*

PUC (Class 12), Science (PCMB), KSEEB, KLE's SCP PU College, Mahalingpur *2023* | *89.16%*

SSLC (Class 10), KSEEB, KLE's SCP High School, Mahalingpur *2021* | *94.84%*